Infrastructure as Code

Software Architecture

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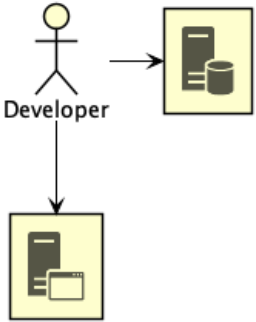
Infrastructure as Code

How did we get here?

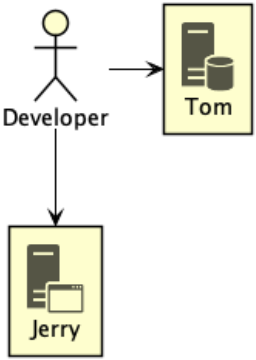
Pre-2000

The *Iron Age*

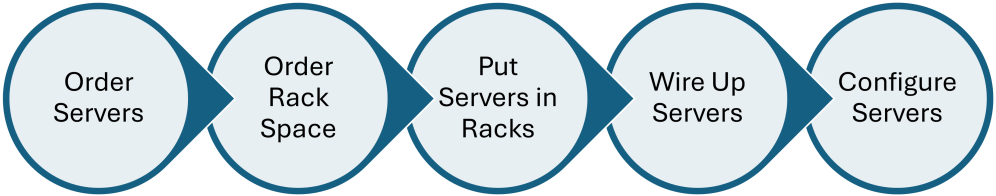
Iron Age



Iron Age



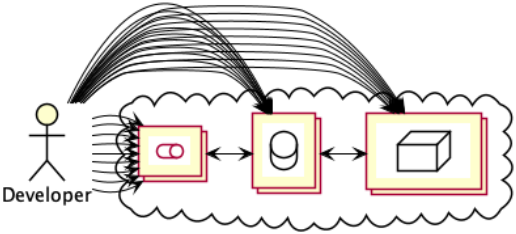
Scaling


Order Servers
Order Rack Space
Put Servers in Racks
Wire Up Servers
Configure Servers

Introducing...

The *Cloud Age*

The Cloud Age



When faced with complexity

Automate it!

The larger story

Server Config Config Management

The larger story

Server Config	Config Management
Application Config	Config Files

The larger story

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Application Config	Config Files
Provisioning	Infrastructure Code

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The larger story

Server Config	Config Management
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Provisioning	Infrastructure Code
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Testing	Automated Tests
Database Administration	Schema Migration
Specifications	Behaviour Driven Development

Definition 0. Infrastructure Code

Code that provisions and manages *infrastructure resources*.

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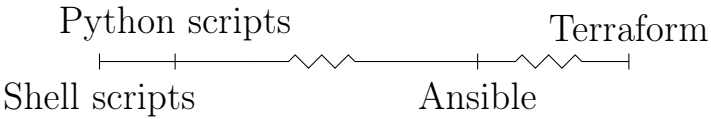
Definition 0. Infrastructure Resources

Compute resources, networking resources, and storage resources.

Infrastructure Code



Shell Scripts

```
#!/bin/bash

SG=$(aws ec2 create-security-group ...)

aws ec2 authorize-security-group-ingress --group-id "$SG"

INST=$(aws ec2 run-instances --security-group-ids "$SG" \
        --instance-type t2.micro)
```

Python

```python
import boto3

def create_instance():
    ec2_client = boto3.client("ec2", region_name="us-east-1")
    response = ec2.create_security_group(...)
    security_group_id = response['GroupId']

    data = ec2.authorize_security_group_ingress(...)

    instance = ec2_client.run_instances(
        SecurityGroups=[security_group_id],
        InstanceType="t2.micro",
        ...
    )
```

Terraform

```
resource "aws_instance" "hextris-server" {
    instance_type = "t2.micro"
    security_groups = [aws_security_group.hextris-server.name]
    ...
}

resource "aws_security_group" "hextris-server" {
    ingress {
        from_port = 80
        to_port = 80
        ...
    }
    ...
}
```

Question

Notice anything different?

The main difference

Imperative vs. Declarative

Declarative IaC

- Define your *desired* infrastructure state
  - as code
- Engine interprets difference between the *desired* and *actual* state
  - Modifying infrastructure to deliver *desired* state

Infrastructure Code

- Provisions and manages *infrastructure resources*.

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Infrastructure Code

- Provisions and manages *infrastructure resources*.
- Only one part of the movement to *automate* the complexities of development.
- Ranges from simple shell scripts up to. . . ?
- Tendency to be *declarative*.

Typo?

Infrastructure Code $\neq$ Infrastructure *as* Code

Definition 0. Infrastructure as Code

Following the same *good coding practices* to manage Infrastructure Code as standard code.

Warning!

Infrastructure as Code still *early* and quite *bad*.

Question

What are *good coding practices*?

Good Coding Practice #1

Everything as Code

```
#!/bin/bash

./download-dependencies
./build-resources
cp -r output/* artifacts/
```

```
#!/bin/bash

./download-dependencies
./build-resources
cp -r output/* artifacts/
```

```
$ cp: directory artifacts does not exist
```

```
resource "aws_instance" "hextris-server" {
    instance_type = "t2.micro"
    security_groups = ["sg-6400"]
    ...
}
```

```
resource "aws_instance" "hextris-server" {
    instance_type = "t2.micro"
    security_groups = [aws_security_group.hextris-server.name]
    ...
}

resource "aws_security_group" "hextris-server" {
    ingress {
        from_port = 80
        to_port = 80
        ...
    }
    ...
}
```

Everything as code avoids

Configuration drift

Configuration drift creates

Snowflakes

Benefits

1. Reproducible

Good Coding Practice #2

Version Control

Benefits

1. Restorable
2. Accountable

Good Coding Practice #3

Automation

Benefits

1. Consistent

Good Coding Practice #4

Code Reuse

Benefits

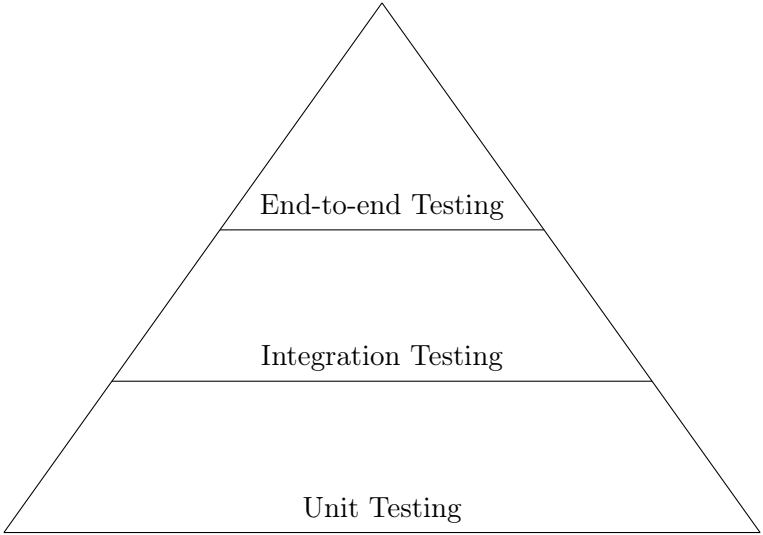
1. Better[1] code
2. Less work
3. Only one place to update (or verify)

[1] generally

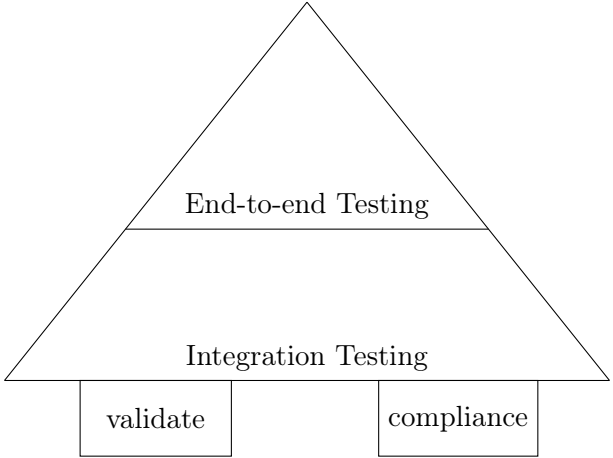
Good Coding Practice #5

Testing

Test Pyramid



IaC Test Pyramid



```go
func TestTerraformAwsInstance(t *testing.T) {
    terraformOptions := terraform.WithDefault(t, &terraform.Options{
        TerraformDir: "../week03/",
    })

    defer terraform.Destroy(t, terraformOptions)
    terraform.InitAndApply(t, terraformOptions)

    publicIp := terraform.Output(t, terraformOptions, "public_ip")
    url := fmt.Sprintf("http://%s:8080", publicIp)

    http_helper.HttpGetWithCustomValidation(t, url, nil, 200,
        func(code, resp) { code == 200 &&
                           strings.Contains(resp, "hextris")})
}
```

```
Feature: Define AWS Security Groups

Scenario: Only selected ports should be publicly open
    Given I have AWS Security Group defined
    When it contains ingress
    Then it must only have tcp protocol and port 22,443 for 0.0.0.0/0
```

Benefits

1. Trust

Prac Next Week

Learn how to use Terraform to write IaC and deploy resources on AWS.